A Spoken Register for the Sky

A proposal for giving stars and planets names that people can actually say

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Summary

Astronomers name the objects they find with reference codes: `Kepler-452b`, `WASP-121b`, `2MASS J05551028+0724255`. These codes work well for astronomers. They work badly for everyone else, because there is nothing in them a listener can hold on to.

This paper suggests we stop trying to fix the codes and add something alongside them instead: a second name for each object, meant for saying out loud. The name would tell you which constellation to look in, and would take its distinctive part from a pool of words meaning “star”, “sky” or “moon”, contributed by languages from all over the world.

Nothing about how astronomy works would change. The reference codes stay exactly as they are. This is simply a list of friendlier names kept alongside them, for the few thousand objects that ordinary people ever hear about.

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1. The problem

About six thousand planets have now been found orbiting other stars. Nearly all of them are known publicly by the code the discovery team gave them.

Those codes are precise and forgettable. Think about what a viewer has to do to remember `Kepler-452b`:

- work out that “Kepler” is a telescope, not a place

- hold on to a four-digit number that means nothing
- notice that the small letter on the end is doing an important job

That is three things to remember, and none of them repays the effort, because the number carries no information you can use. There is nothing to picture and nothing to work out. You could not repeat it to a friend the next day.

You can see the effect in how these objects get talked about. Presenters either skip the name altogether and say “a planet in the constellation Cygnus”, or they read the code out and watch the audience drift. Neither is good. The night sky is the one thing every human being has looked at, and we have managed to make it sound like a stock list.

2. Why the names are like this

The current system is not careless. It solves real problems, and any replacement would have to solve them too.

There are too many objects for names. The Gaia space telescope has catalogued nearly two billion stars. Nobody has two billion memorable names, and no committee could invent them.

Codes must never change. A code has to point at one object and one object only, permanently, across every scientific paper ever written about it. Renaming things breaks the record.

Some codes are instructions. `2MASS J05551028+0724255` looks like noise but is actually a position in the sky, precise to a fraction of a hair’s breadth. It is ugly because it is useful: it tells a telescope exactly where to point.

Meaningful names go wrong. Astronomy has been caught out by this before. A “planetary nebula” has nothing to do with planets. The “Great Attractor” claims more than anyone actually knows. A meaningless number, whatever else you say about it, can never turn out to be false.

Naming has a bad history. For a long time, European names were stamped on everything anyone found, anywhere. The International Astronomical Union, which oversees naming, is deliberately slow and careful about this now. That caution is a feature, not an obstacle.

Any suggestion that ignores these five points deserves to be ignored in turn.

3. The idea

Here is the shift in thinking the whole proposal rests on.

Almost every organisation keeps two kinds of name for the same thing. A hospital gives you a patient number; the staff still call you by your name. A shop's stock system has a barcode for a jacket; the label still says "navy wool coat". The number is for the machine and must never change. The name is for people and can be whatever helps them.

Astronomy has built an excellent set of numbers and never got round to the names. So presenters end up reading the numbers out on television, because there is nothing else to read.

The proposal is not to rename anything. It is to add the missing half.

4. The rules I set myself

1. **Add, never replace.** The existing codes remain the official identifier. This is a second name kept alongside. No scientific paper has to change.
2. **Respect names that already exist.** Where an object already has a real name — given by the IAU, or handed down in Arabic, Greek, Polynesian or any other tradition — that name wins. The list records it and invents nothing.
3. **Say where it is.** The one useful fact about an object that ordinary people can act on is which constellation it sits in. Constellation boundaries are fixed, and they do not shift when someone takes a better measurement. So that goes in the name.
4. **Keep the science out of the name.** How an object is classified changes as instruments improve. Something described today as a giant planet might be reclassified tomorrow. That must not force a rename.
5. **No culture gets to go first.** No gods, no heroes, no explorers, no nations. Just the plainest possible word.
6. **Nobody chooses.** Which language goes with which star is decided by a calculation, not by a person or a committee.
7. **Once given, never changed.** A name is worked out once and then written down. It is never worked out again.

5. How a name is built

A name has two parts, plus a description that travels with it but is not part of it.

Part one — where it is

The standard three-letter abbreviation for the constellation. These already exist and astronomers already use them every day: `Cyg` for Cygnus, `Peg` for Pegasus, `Aqr` for Aquarius, `Dor` for Dorado.

Using them costs nothing, because nobody has to agree to anything new. They are Latin, which does slightly undercut the argument about not favouring any one culture. That is a genuine compromise rather than a solved problem — but they are an existing shared standard rather than something freshly imposed, and they buy instant familiarity with the people who would have to adopt this.

Part two — the distinctive word

A word taken from a shared pool, chosen by calculation from the object's reference code.

Here is the rule that holds the pool together: every word in it means "star", "sky" or "moon" in some human language.

That is the answer to the fairness problem. Every culture on Earth has looked up and named what it saw. The pool is built from those plain, ordinary nouns — *nyota* in Swahili, *bituin* in Tagalog, *quyllur* in Quechua, *seren* in Welsh, *whetū* in Māori, *ulloriaq* in Inuktitut, *noquisi* in Cherokee, *biddéw* in Wolof, *kintana* in Malagasy.

No one's gods are pushed onto anyone else. No country's explorers are commemorated. The sky is named, in every language, after the sky.

Which language each word came from is recorded next to it, so the origin stays visible and can be credited rather than quietly lost.

Part three — which planet

A star and its planets sit in the same place and share the same word from the pool: the word is given to the whole system, not to each object separately. A final vowel is what tells them apart.

The star gets no vowel. Each planet gets one, matching the letter astronomers already use:

ASTRONOMERS' LETTER	ENDING
b	-a
c	-e
d	-i
e	-o
f	-u
g	-ae
h	-ea

So the Kepler-452 system reads:

- Kepler-452, the star → **Cyg Stjarna**
- Kepler-452 b, the planet → **Cyg Stjarna-a**

And separately: what kind of thing it is

Every entry also carries a short description — *the rocky world*, *the scorching giant*, *the pulsar world*. This is what a presenter would say just before the name: "the rocky world Aqr Quyllur-o". It works the same way as saying "the physicist Marie Curie". The word "physicist" tells you what sort of person is coming next. It is not part of her name, and calling her a chemist instead would not change what she is called.

This distinction does real work, and it is worth showing why.

Suppose we had built the description into the name instead. WASP-12b is currently classed as a scorching giant planet, so we might have called it `Aur Hotgiant-a`. If later measurements reclassified it — and the line between a very large planet and a very small failed star is genuinely blurry, with objects having crossed it before — the name would now be telling people something untrue. Fixing it would mean issuing a replacement, and every article, broadcast and reference using the old one would be wrong.

Keeping the description outside the name makes that correction cost nothing. The name `Aur Yildiz-a` does not move. Only the spoken description changes, from "the scorching giant" to "the failed star".

6. Examples

These are the actual results from the working version, not hand-picked to look good.

REFERENCE CODE	CONSTELLATION	NEW NAME	WHERE THE WORD COMES FROM	SAID AS
Kepler-452 b	Cygnus	Cyg Stjarna-a	star, Icelandic	“the super-Earth Cyg Stjarna-a”
TRAPPIST-1 e	Aquarius	Aqr Quyllur-o	star, Quechua	“the rocky world Aqr Quyllur-o”
Proxima Centauri b	Centaurus	Cen Seren-a	star, Welsh	“the nearest world Cen Seren-a”
TOI-700 d	Dorado	Dor Noquisi-i	star, Cherokee	“the rocky world Dor Noquisi-i”
51 Pegasi b	Pegasus	*Dimidium*	—	already has a name; kept

That last row matters as much as the others. Where a name already exists, the list records it and steps aside.

7. Who decides, and how to take part

The list is a public file that anyone can read, copy or correct. There are two ways to contribute.

Adding a word. Anyone can suggest a word for the pool, giving the language, the meaning, how it is written and roughly how it is said. Words are checked before they go in. This is not politeness for its own sake: a word that means “star” in a dictionary can mean something unfortunate in a neighbouring dialect, and only people who speak the language will catch that.

Adding an object. Anyone can submit a reference code and its constellation. The calculation produces the name; no person picks it.

Two rules keep the list trustworthy:

- **Names are only ever added, never taken back.** Once given, a name stays given.
- **No language may dominate.** If the calculation lands on a word already used in that constellation, it moves along to the next free one. Over the whole list this spreads the languages evenly.

None of this needs anyone’s permission to be useful. It is a list, published openly, that anyone may use. If it is good, people will use it.

8. What’s wrong with it so far

Some constellations will run out of words. The Kepler telescope alone dropped thousands of planets into Cygnus. A pool of fifty words will not stretch that far. It needs to grow into the hundreds — which is achievable, since there are roughly seven thousand living languages — and the busiest constellations may eventually need two words rather than one. Clumsy, but only where the crowding demands it.

The Latin abbreviations. Discussed in part five. A reasonable trade, but a trade all the same.

Checking words takes time. Getting speakers to confirm each word is the real bottleneck, and it should be. A list that grows slowly and correctly is worth far more than one that grows quickly and offends people.

Getting anyone to use it. This is the hardest problem and it is not a technical one.

9. How this could catch on

The whole sky is not the target. The target is the handful of objects that actually turn up in documentaries, news stories and museum captions — perhaps a couple of hundred, growing slowly.

1. Name those properly, with a checked pool of words.
2. Publish the list openly, along with the tool that produces the names.
3. Get one person who talks to the public — a presenter, a planetarium, a science journalist — to use it.

One television producer saying these names on air would do more than any committee’s approval. That is the entire strategy.

Appendix A — How a name is picked

The reference code is put through a standard calculation that turns text into a number. That number decides which word in the pool to use. If the word is already taken in that constellation, it steps along to the next free one. The result is then written down and never worked out again.

```
To name an object:

    If it already has a proper name, record that and stop.

    Take the constellation's three-letter abbreviation.
    Turn the reference code into a number, and use that
      number to pick a word from the pool.
    If that word is already used in that constellation,
      step to the next word, and keep stepping until one is free.
    Add the vowel that matches the planet's letter.

    Write the finished name down permanently.
```

The calculation is run **once**, when the name is first given. The written-down name is what counts from then on. It is never worked out again from the reference code, because reference codes do occasionally get merged, split or reclassified, and a name that quietly changes when that happens is not a name at all.

Appendix B — What has been built

A working version accompanies this paper: the list itself (`register.json`), a small program that adds names to it (`issue.py`), and a web page that displays it (`index.html`). Constellations are not typed in by hand: each host star's position is checked against the official IAU boundaries, so that part of every name is derived rather than guessed. The list currently holds 179 objects, using words from 44 languages across 42 constellations. Twenty of those already carry a proper name and are recorded rather than renamed.

The program enforces the “never changed” rule directly. Names are added to the end of the list and the list is never rebuilt from scratch. A checking command looks for duplicates, unknown constellations and clashes.

The words currently in the pool were gathered from reference sources and **have not yet been confirmed by native speakers**. Every entry has a field recording who vouched for it, and every one of those fields is currently empty. This is the most useful thing anyone reading could help with.

Everything is at **github.com/matthewsantley/spoken-register**. Correspondence to the address at the top.

Appendix C — What this does not do

- It does not rename anything in the scientific record.
- It does not need the IAU’s approval, though it would be glad of it.
- It does not try to name the two billion objects in the Gaia catalogue.
- It does not encode distance, mass, temperature or anything else that gets measured.
- It does not claim these names are beautiful. It claims they can be said.